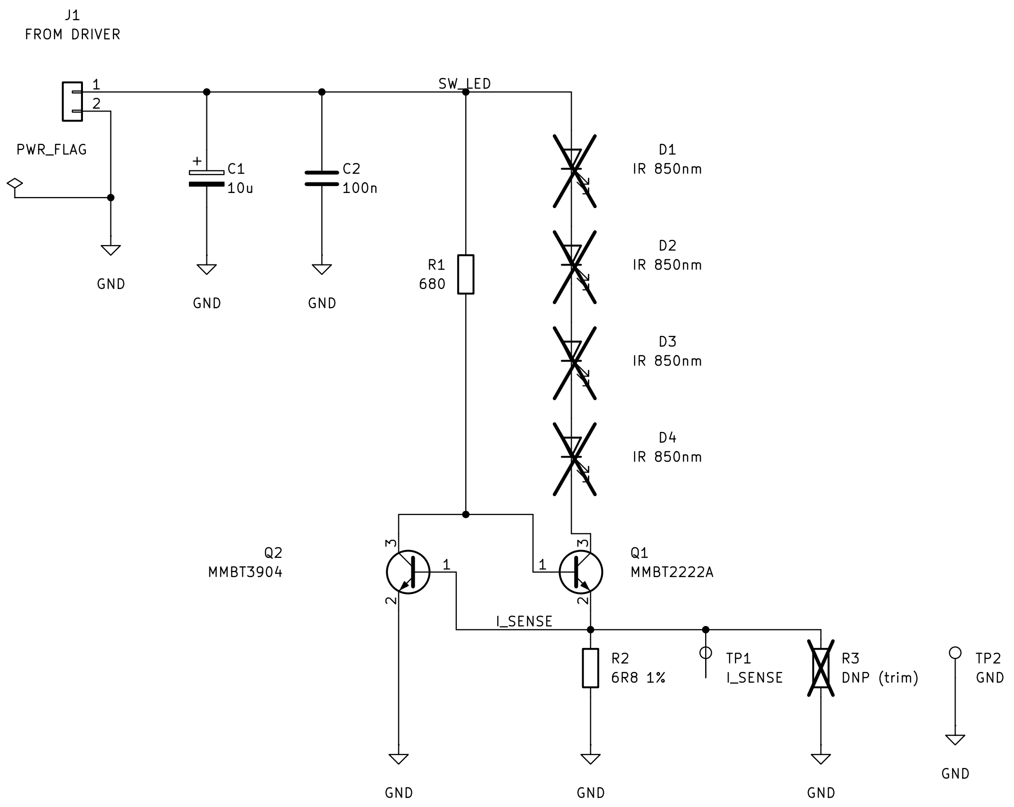

MARKER BOARD — build 4 (A, B, C, D), all identical

Q1/Q2/R2 is a constant—current sink: Q2 steals Q1's base drive as soon as R2 reaches a Vbe, so

I_LED = Vbe(Q2) / R2 ~= 0.65 / R2

independent of the supply, of C1's droop during the pulse, and of Q1's beta. That is the point: all four markers land on the same current without matching LEDs, so ONE camera exposure targets every marker at once, and a sagging battery dims nothing until the sink drops out.

```
R2 SELECTION      4R3 -> 151 mA     6R8 -> 96 mA (fitted)
                  8R2 ->  79 mA     13R ->  50 mA
```

R3 is a DNP footprint in parallel with R2 for fine trim upward.
Measure across TP1—TP2 with a scope during a pulse: I = V / R2.

96 mA is 2.5x the 38.2 mA DC point from LED_characterisation.xlsx, at 0.09% duty — far inside any datasheet pulse rating. Q1 dissipates ~0.33 W DURING the pulse and 0.3 mW average.

```
HEADROOM BUDGET at 96 mA (this is what sets the 12 V rail):
   4 x Vf(850nm @ 96 mA)  ~7.4      D1 Schottky on driver   0.45
   R2 sense                0.65     Q1 Vce, min to regulate 0.50
   Q1 Vds + wiring         0.06     C1 droop                0.31
   --------------------------------------------------------------
   minimum pack voltage   ~9.3 V     <- 8xAA or 3S LiPo, NOT a 9 V PP3
```

D1..D4 FOOTPRINT IS STILL BLANK. DNP excludes them from assembly, NOT from the PCB — the pads must exist, so the package is needed before layout. LED_characterisation.xlsx ranks Kingbright 25 mm best (sigma 0.015 px at 1186 us, vs XL's 6057 us) but records no MPN — fill in the MPN field. Its only failing was fill 0.82 against the 0.85 gate, which Test G (thicker wall) is meant to fix.

LED RING: space D1..D4 as widely as the 20/25 mm shell allows and aim them TANGENTIALLY, not radially — NOTES.md attributes the ring artefact to beam angle, not diffuser diameter.

TWO DIFFERENT MEANINGS OF DNP ON THIS SHEET — do not conflate them:
- D1..D4 ARE FITTED. Marked DNP only so a turnkey assembler skips them; the Kingbright emitters are hand—soldered after assembly so the diffuser standoff can be set by hand. The board is useless without them. The FOOTPRINT is still required for layout.
- R3 IS NOT FITTED. Bare pads, an optional trim in parallel with R2.

UCT MSc — Shannon Pitman

Sheet: /

File: WandMarker.kicad_sch

Title: Wand marker — 4x 850 nm IR LED + constant—current sink

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.4		Id: 1/1